October 27, 29, 2026

UNIT 2: BONDING

G. Chemistry of Selected Main Group Elements
H. Acid Strength and Structure (Chapters 14, 15)

**Worksheet on p-Block Elements

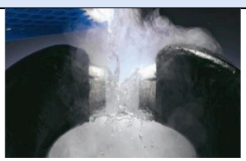
UNIT 3: GASES, LIQUIDS, and SOLIDS

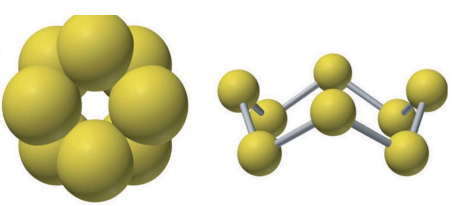
Utilize different CHEM 103 learning outcomes to predict reactivity and physical properties of main group elements and compounds.

GROUP 16 ELEMENTS (CHALCOGENS)

Oxygen and sulfur

Properties. Oxygen forms two allotropes, whereas sulfur forms an extraordinary number of allotropes in addition to S_8:

$O_2(g)$, dioxygen	$O_3(g)$, ozone	$S_8(s)$, (rhombic) sulfur
		
colourless, paramagnetic gas	"bluish?" pungent gas	yellow solid
bp –183 °C	bp –112 °C	mp ~ 120 °C
21 vol % in atmosphere	~8 ppm in stratosphere	crown-shaped molecule



At room temperature, the sulfur molecule is a crown-shaped ring of eight atoms. The most stable S allotrope is orthorhombic α-S_8, which consists of cyclo-S_8.

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Occurrence and uses

Oxygen is the most abundant element on the earth's surface. It is readily available in the atmosphere and is isolated by fractional distillation of liquid air. It has many industrial uses (e.g., steel production, chemical manufacture, sewage treatment, oxidizer for rocket fuels).



The earth's crust contains element sulfur and various sulfide minerals (e.g., FeS_2). H_2S and organosulfur compounds are significant impurities in natural gas and crude oil (especially in Alberta), making it difficult to process and refine. What happens when you burn "sour gas" or dirty gasoline?



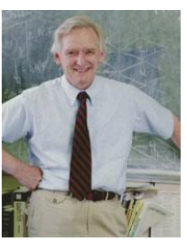
Industrial sources produce the sulfur oxides that lead to acid rain.

Almost all sulfur extracted is destined for preparing sulfuric acid, the world's most important industrial chemical (used as a cheap strong acid in chemical manufacture).

Demo: What happens when you pour sulfuric acid on sugar, $C_{12}H_{22}O_{11}(s)$? https://www.youtube.com/watch?v=m_pProDYqTE

GROUP 18 ELEMENTS (NOBLE GASES)

All are monoatomic gases with extremely low mp and bp (e.g., 4 K for He). With completely filled valence shells ($ns^2\ np^6$) and very high IE, these elements are highly unreactive.



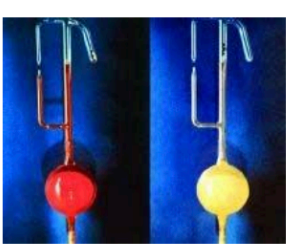
In 1962, Neil Bartlett at UBC was studying the use of PtF_6 as a strong oxidizing agent:

$O_2 + PtF_6 \rightarrow$

He compared the ionization energies of O_2 and Xe:

What if Xe can also be oxidized? From this reasoning, he prepared $XePtF_6$, the first noble-gas compound, as a yellow-orange solid:

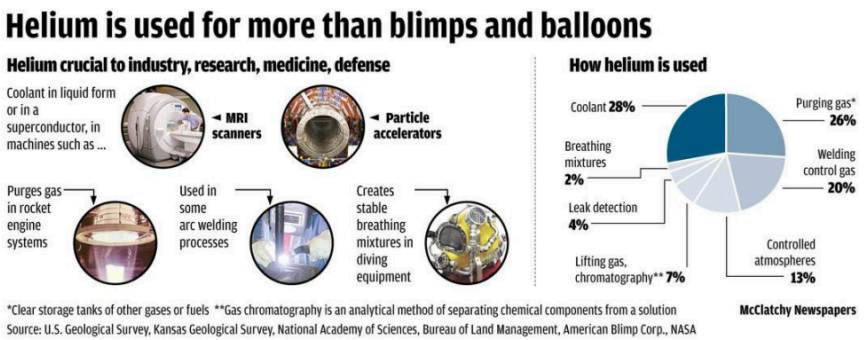
$PtF_6(g)$



$XePtF_6(s)$
Yellow-orange solid

Several fluorides and oxides (XeF_2, XeF_4, XeO_4) are also now known.

Uses. He is an important cryogen, and Ne is used in (what else?) neon signs.



https://www.cbsnews.com/news/helium-critical-material-energy-department-supply-chain/

H. Acid Strength and Structure (14.1 - 14.4, 15.2)

1. Definitions

Arrhenius Definition: acids have H in their formula and form H_3O^+ in aqueous solution. Bases have OH in their formula and form OH^- in aqueous solutions.

Examples: HCl, NaOH

Limitations:

– only applies to aqueous solutions

– definition only includes bases containing OH^- ion

Bronsted-Lowry Definition

acid = proton donor (acid must contain H)

base = proton acceptor (base must contain lone pair to bind H+)

examples: acids = HCl, NH_4^+ and bases = NH_3, NO_2^-

Examples: anything previously identified as an Arrhenius Acid or Base, but with expanded bases (NH_3, F^- =weak base conjugates)

Lewis Definition (Section 15.2)

acid = electron pair acceptor (generally have vacant orbitals to accommodate electrons)

base = electron pair donor (must have electron pairs to donate)

Includes all Bronsted-Lowry acids and bases and includes some new acids because the definition does not require presence of proton.

- The Lewis definition views an acid-base reaction as the ***donation and acceptance of an electron pair*** to form a ***covalent bond***.



- A Lewis base must have a ***lone pair of electrons*** to donate.
  - Any substance that is a Brønsted-Lowry base is also a Lewis base.

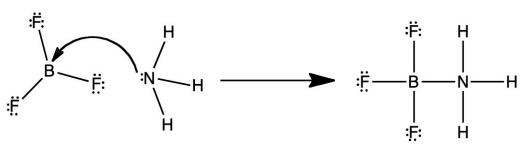
- A Lewis acid must have a ***vacant orbital*** (or be able to rearrange its bonds to form one) to accept a lone pair and form a new bond.
  -Many substances that are **not** Brønsted-Lowry acids are Lewis acids.

- The Lewis definition expands the classes of ***acids***.

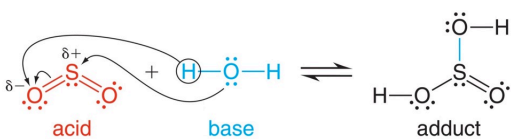
Examples of Lewis acids

(1) Molecules with **electron deficient atoms**

Write the neutralization reaction between $BF_3(g)$ and $NH_3(g)$, to give a Lewis acid-base adduct:

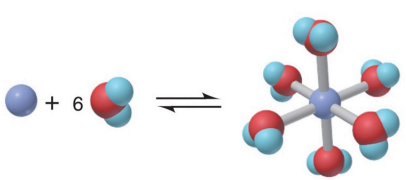


(2) Molecules that contain a **polar multiple bond** often function as Lewis acids



(3) Formation of **coordination complexes**:

$Al^{3+} + 6\ H_2O \rightarrow$



metal cation acts as a Lewis acid

The O atom of an H_2O molecule donates a lone pair to an available orbital on the metal cation.

Metal cations are Lewis acids since they are able to accept electrons. (e.g., Cu^{2+}, $Fe^{2+,}$ $Fe^{3+)}$

An atom, ion, or molecule with an incomplete octet of electrons can act as an Lewis acid (e.g., BF_3, AlF_3).

Molecules where the central atom can have more than 8 valence shell electrons can be electron acceptors, and thus are classified as Lewis acids (e.g., $SiBr_4$, SiF_4).

Molecules that have multiple bonds between two atoms of different electronegativities can act as Lewis acids (e.g., CO_2, SO_2)

An atom, ion, or molecule with a lone-pair of electrons can be a Lewis base. Each of the following anions can "give up" their electrons to an acid, e.g., OH^-, CN^-, CH_3COO^-, NH_3, H_2O, CO

Identify the Lewis acids and bases

$Fe^{3+} + 2H_2O \rightleftharpoons Fe(OH)^{2+} + H_3O^+$

$4CO + Ni \rightleftharpoons Ni(CO)_4$

$H_2SO_4 + NH_3 \rightleftharpoons HSO_4^- + NH_4^+$

$CN^- + H_2O \rightleftharpoons HCN + OH^-$

Bronsted-Lowry Definition

acid = proton donor (acid must contain H)

base = proton acceptor (base must contain lone pair to bind H+)

examples: acids = HCl, NH_4^+ and bases = NH_3, NO_2^-

In an acid-base reaction, a proton is transferred from the acid to the base

$$HA + B \rightleftarrows BH^+ + A^-$$

Write the reaction (in the gas phase) between HCl(g) and NH_3(g):

More typical examples involve reaction with the solvent:

$NH_3(g) + H_2O(l) \rightarrow$

$NH_4^+(aq) + H_2O(l) \rightarrow$

H_3O^+ is called the hydronium ion (is equivalent to H^+).

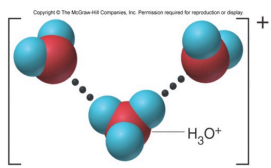
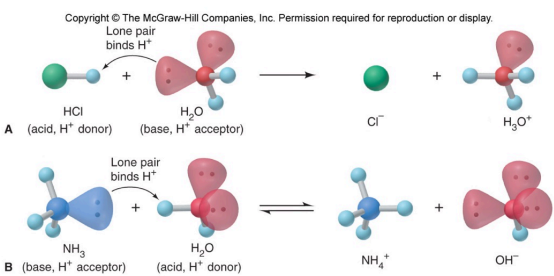


Figure 16.1



Note that water can act as either an acid or a base in a Brønsted-Lowry acid-base reaction.

Figure 16.2

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Example: Identify the acid and base, and indicate which definitions apply.

(a) $CaO(s) + SO_2(g) \rightarrow CaSO_3(s)$

(b) $OCl^-(aq) + H_2O(l) \rightleftharpoons HOCl(aq) + OH^-(aq)$

(c) $HCl(aq) + NaOH(aq) \rightarrow NaCl + H_2O$

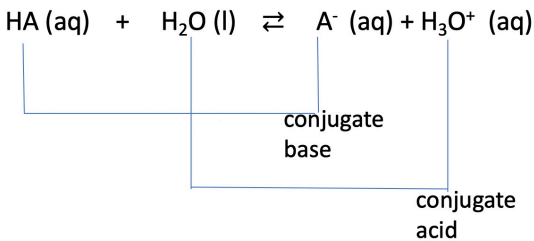
Amphoteric Species: can react as either an acid or a base.

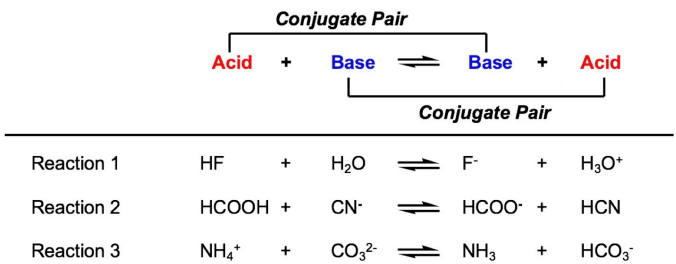
--*Review basic oxides and acidic oxides (Unit 1).* Some oxides can be amphoteric: BeO, Al_2O_3

--Intermediate species (related to polyprotic acids) are amphoteric (not yet covered).

2. Bronsted-Lowry Acids/Bases: conjugates and strength

Conjugate pair = two substances related to each other by gain or loss of a proton.





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Measures of Acid and Base Strength (14.3)

In aqueous solution, we quantify the strength of an acid or a base by relating to the magnitude of the equilibrium constant *K*.

Large K_a = stronger acid, small K_a = weaker acid.

Acid, HA

$HA(aq) + H_2O(l) \rightarrow A^-(aq) + H_3O^+(aq)$

acid ionization constant:

$$K_a = \frac{[A^-][H_3O^+]}{[HA]}$$

Base, B

$B(aq) + H_2O(l) \rightarrow BH^+(aq) + OH^-(aq)$

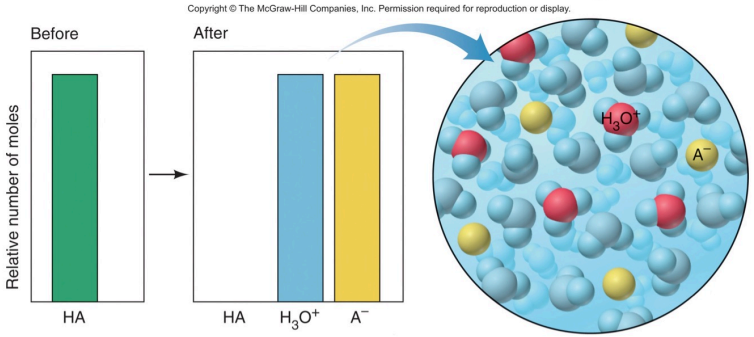
base ionization constant:

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

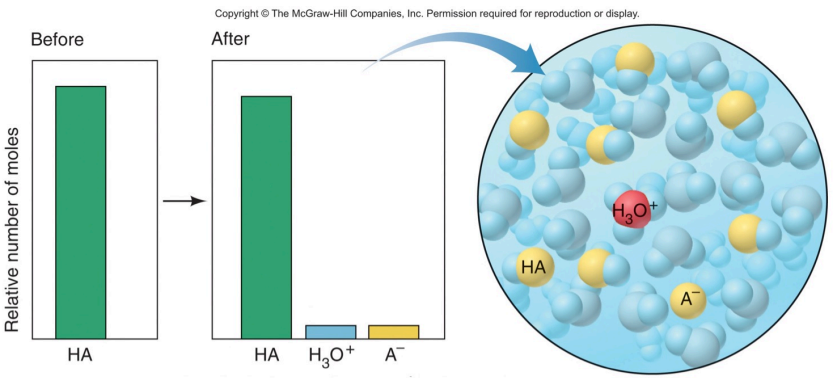
Examples of K_a (aq) for various acids:

HNO_3, nitric acid	very large
HSO_4^-, hydrogen sulfate ion	1.1×10^{-2}
CH_3COOH, acetic acid	1.8×10^{-5}
HCN, hydrocyanic acid	6.2×10^{-10}
H_2O, water	1.0×10^{-14}

Figure 16.4 The extent of dissociation for strong acids.

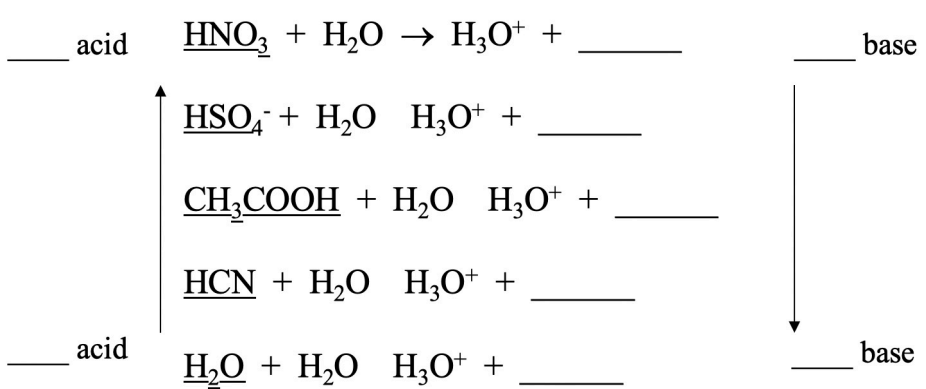


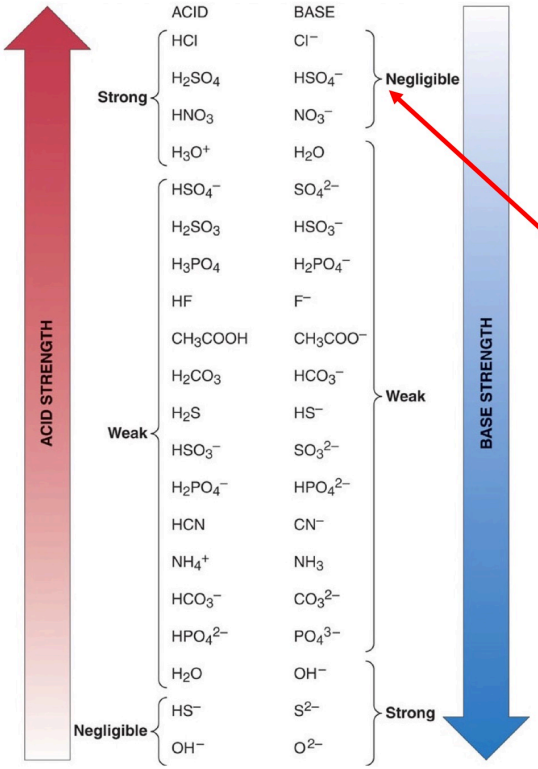
The extent of dissociation for weak acids.



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Since acids and bases come in conjugate pairs, we can also deduce the relative strengths of the conjugate bases:





- **Memorize the typical strong acids (see supplementary handout on eClass)** (learn to recognize).
- **The stronger the acid, the weaker the conjugate base.**
- **Conjugates to strong acids are so weak that their strength is negligible.**

Figure 16.3

Strengths of conjugate acid-base pairs.

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Strong Acids and Bases (MEMORIZE)

Strong Acids		Strong Bases
HCl	hydrochloric acid	LiOH
HBr	hydrobromic acid	NaOH
HI	hydroiodic acid	KOH
$HClO_4$	perchloric acid	RbOH
HNO_3	nitric acid	CsOH
H_2SO_4	sulfuric acid	$Mg(OH)_2$
		$Ca(OH)_2$
		$Sr(OH)_2$
		$Ba(OH)_2$

Acids and Bases Worksheet

Definitions:
Arrhenius acids and bases dissociate to liberate H^+ and OH^-, respectively.
Bronsted-Lowry acids are proton donors and Bronsted-Lowry bases are proton acceptors.
Lewis acids are electron pair acceptors and Lewis bases are electron pair donors.

1. Strong Acids and Bases (MEMORIZE)

Strong Acids		Strong Bases
HCl	hydrochloric acid	LiOH
HBr	hydrobromic acid	NaOH
HI	hydroiodic acid	KOH
$HClO_4$	perchloric acid	RbOH
HNO_3	nitric acid	CsOH
H_2SO_4	sulfuric acid	$Mg(OH)_2$
		$Ca(OH)_2$
		$Sr(OH)_2$
		$Ba(OH)_2$

Strong Acids/Bases: large equilibrium constants; dissociate completely in solution.
Anions from strong acids: Cl^-, NO_3^- or **Cations from strong bases**, Na^+, K^+ have essentially **no acid-base properties** in water since they are too weak to display significant acid-base properties.

Strong acids include:

- hydrohalic acids (HCl, HBr, and HI)
- oxoacids in which the number of O atoms exceeds the number of ionizable protons by two or more (eg., HNO_3, H_2SO_4, $HClO_4$.)

Strong bases include:

- water-soluble compounds containing O^{2-} or OH^- ions with cations from those of the most active metals (Group 1 or Group 2).

e.g., M_2O or MOH (Group 1) and MO or $M(OH)_2$ (Group 2).

Weak acids and Bases: small or intermediate equilibrium constants; do not dissociate completely
Conjugates to weak acids are weak bases; the conjugates to weak bases are weak acids.

Examples:

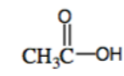
Inorganic acids: HF (hydrofluoric acid), HCN (hydrocyanic acid), HClO (hypochlorous acid), HNO_2 (nitrous acid), H_3PO_4 (phosphoric acid).

Inorganic bases: From the above weak acids, we can obtain the conjugate weak bases: F^- (fluoride ion), CN^- (cyanide ion), OCl^- (hypochlorite ion), NO_2^- (nitrite ion).

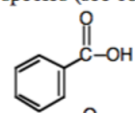
Organic examples:

Acids: examples: acids with a carboxylic acid group R-**COOH**, where R is the abbreviation for the rest of the molecule), or positively charged ammonium species (see conjugates to the weak base examples below)

carboxylic acids (weak acids)

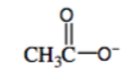


acetic acid

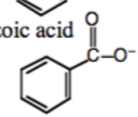


benzoic acid

carboxylate anions (conjugate weak bases)



acetate anion



benzoate anion

Bases: examples: amines (see below) or anionic conjugates of carboxylic acids (R-COO^-) (see conjugates to weak acids above)

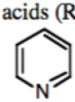
Amines (weak bases)



ammonia



methylamine

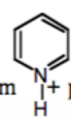


pyridine

Ammonium ions (weak acids)

ammonium

methylammonium



pyridinium

Weak acids include:

- the hydrohalic acid HF,
- acids in which H is not bonded to O or to a halogen (eg., HCN),
- oxoacids in which the number of O atoms equals or exceeds the number of ionizable protons by one (eg., HClO, HNO_2), and
- carboxylic acids, which have the general formula RCOO**H** (eg., CH_3COOH and C_6H_5COOH.)

Weak bases include

- ammonia (NH_3),
- amines, which have the general formula:

$$R\ddot{N}H_2,\ R_2\ddot{N}H,\ \text{or}\ R_3\ddot{N}$$

The common structural feature is an N atom with a lone electron pair.

3. Salts (14.4)

- ionic compounds that result from an acid-base *neutralization* reaction:

$$HCl + NaOH \rightarrow _____ + H_2O$$

- These salts can be isolated as solids. If they are dissolved, the ions themselves may possess acid-base properties (because they originated as partners of conjugate acid-base pairs).

- When a salt dissolves, one or both of its ions may react with water and affect the pH of the solution.

  Examples: $NaCl$, K_2CO_3, $Al(NO_3)_3$

  Will their solutions be acidic, basic, or neutral?

Cations

- Cations derived from **strong bases** (e.g., Na^+, Ca^{2+}) do not react with water: **no effect**.

- All other cations react with water to yield weakly acidic solutions.
  Examples: NH_4^+, transition metal ions, Al^{3+} → **ALL Weakly Acidic**

Anions

- Anions derived from **strong acids** (e.g., Cl^-, I^-, Br^-, NO_3^-, ClO_4^-) do not react with water: **no effect**.

- All other anions react with water to yield weakly basic solutions.
  Examples: F^-, CH_3COO^- → **ALL Weakly Basic**

NaCl

K_2CO_3

$Al(NO_3)_3$

Recognizing Acids and Bases

Identify whether the following would give acidic, basic, or neutral solutions (when dissolved in water):

$CaCl_2$

NaF

NaCl

$(CH_3)_2NH$

KBr

NH_4I

KCN

Na_2CO_3

$KClO_4$

CH_3COOH

Strong Acids and Bases (MEMORIZE)

Strong Acids		Strong Bases
HCl	hydrochloric acid	LiOH
HBr	hydrobromic acid	NaOH
HI	hydroiodic acid	KOH
$HClO_4$	perchloric acid	RbOH
HNO_3	nitric acid	CsOH
H_2SO_4	sulfuric acid	$Mg(OH)_2$
		$Ca(OH)_2$
		$Sr(OH)_2$
		$Ba(OH)_2$

4. Molecular Structure and Acid Strength (Sec 14.3)

(1) Binary acids, H–X

The acidic proton is bonded directly to a nonmetal atom X. The weaker and more polar the H–X bond, the easier it is to dissociate H^+ and the stronger the acid.

$$H\text{–}X \rightarrow H^+ + X^-$$

Down a group, consider bond length. The longer the bond, the weaker the bond and the stronger the acid.

	Length (pm)		K_a
H-F	92	weak	6.6×10^{-4}
H-Cl	127	strong	1.3×10^{6}
H-Br	141	strong	10^{8}
H-I	161	strong	10^{9}

****Down a group, acid strength increases.**

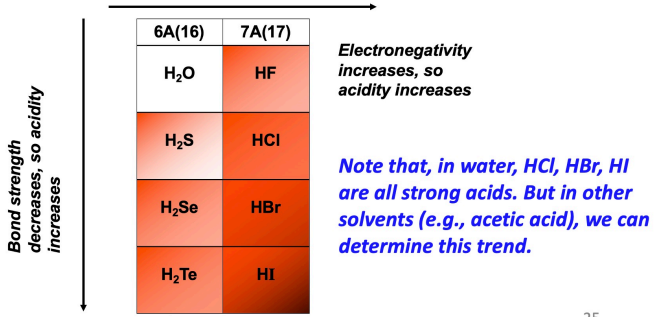
Across a period, consider bond polarity (ΔEN): the more polar the H-X bond, the stronger the acid.

e.g.		CH_4	NH_3	H_2O	HF
	ΔEN	0.4	0.9	1.4	1.9

****Across a period, acid strength increases**

Figure 16.11 The effect of atomic and molecular properties on nonmetal hydride acidity.

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(2) Oxyacids= compounds where acidic proton is bound to oxygen.

e.g., H_2SO_4, H_2SO_3, HNO_3, HNO_2

O-H bond is common to all these, so must consider other factors.

- **electronegativity of the central nonmetal (E)**
- **number of terminal O atoms present.**

Oxyacids with *same number of oxygens* around "E" acid strength increases with electronegativity of E.

The more EN E is, the more electron density is drawn away from the O-H bond. Polarizes O-H bond, H^+ is lost easier = stronger acid.

e.g.,	H-O-I	H-O-Br	H-O-Cl
K_a	2.3×10^{-11}	2.3×10^{-9}	2.9×10^{-8}

Oxyacids with different number of terminal oxygens around E: acid strength increases with number of O atoms.

Oxygens are electronegative and draw electron density away from E, which makes O-H bond more polar. The more O atoms, the greater the effect and the more easily the H^+ is lost = stronger acid.

e.g.,	$HOCl$	$HClO_2$	$HClO_3$	$HClO_4$
K_a	2.9×10^{-8}	1.1×10^{-2}	1	$>10^7$

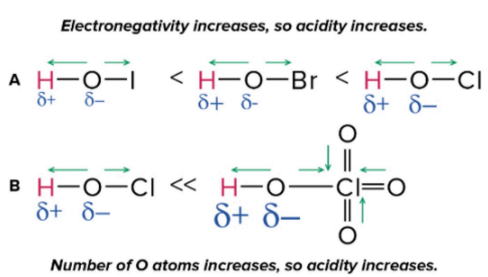
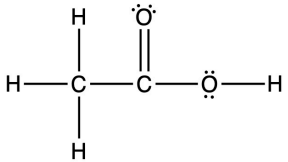


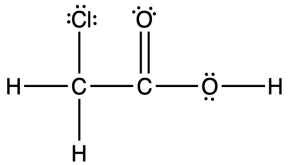
Figure 16.12

(3) Organic acids

EN of other atoms present will affect acid strength:



versus



$K_a = 10^{-5}$

$K_a = 10^{-3}$ (stronger)

Which is the stronger acid?

HNO_3 or $HClO_4$

Which is the stronger acid?

$HClO_3$ or H_2SO_3

Which is the stronger acid?

